



Regular Expressions

1. Introduction to Regular Expressions
2. Creating regular expression
3. Syntax and rules
4. Pattern matching
5. Data validation

- Its a java script object
- It defines patterns used to match character combinations in a string
- Mainly used for searching within a string or replacing paticular substring in a string

- Can be created using 2 ways
 - Using a regular expression literal
 - Syntax:
 - `var pattern = /pattern/modifiers`
 - Using constructor function of the RegExp object
 - Syntax:
 - `var pattern = new RegExp(pattern,modifiers)`

- Regular expressions can contain letters, digits, and various special characters that represent matching rules
- Pattern modifiers(flags)
 - i: indicates case insensitive search.
 - g: specifies that searching should not be stopped after first find
- Symbols and special characters
 - caret (^)
 - Used for specifying pattern to be searched at the beginning
 - dollar(\$)
 - Used for specifying pattern to be searched at the end
 - Square brackets []
 - Used to define a character set from which any of the character is searched

- Square brackets with caret [^]
 - Used to specify match any character other than given the character set
- Quantifiers
 - Used to specify how many times a particular element occur
 - i. * : zero or more occurrence
 - ii. + : one or more occurrence
 - iii. ? : zero or one occurrence
 - iv. {n} : exact n occurrences
 - v. {n, m} : occurrences from n to m
 - vi. {n,}: minimum n occurrences, no limit on upper side

- Parentheses
 - Allows to group parts of regexp pattern. Symbol is ()
- Special codes
 - represent specific characters or patterns
 - i. \d: any digit
 - ii. \D: any character but not digit
 - iii. \s: single whitespace
 - iv. \S: single nonwhite space
 - v. \w: word character
 - vi. \W: nonword character
- Special character
 - Characters with spacial meaning
 - i. . : matches any character except newline
 - ii. | : used for alteration
 - iii. \ : to remove the special meaning of a character

- Several functions in RegExp and String allow to match the pattern for searching, replacing, splitting.
- RegExp functions
 - test: argument is string in which pattern to be searched and returns boolean value
 - exec: returns array containing information of match
- String functions
 - match: returns an array of all the matching strings
 - replace: replaces matched string with new string
 - search: search is a powerful function which takes regexp as a argument
 - split: allows to split on pattern matched on regexp.

- Examples of regular expressions
 - Validating string containing only small alphabets with minimum length as 2 : `/[a-z]{2,}/`
 - Validating phone number with 10 digits : `/[\d]{10}/`
 - Validating name with first capital character and rest minimum 2 small alphabets: `/[A-Z][a-z]{2,}/`
 - Validating email id :
`/^\w{6,}\@\w{4,}\.\w{2,}$/`

- Before submitting data to server side, data can be validated against a regexp pattern.
- If the data found is invalid, we can accept new set of data
- This can even be visually managed in the HTML form by changing the background or applying some border to form field.
- Data can be submitted only after all the form fields are found valid.